CYBERIUM

NETWORKING FOR ETHICAL HACKERS



INTRODUCTION

A comprehensive networking course is indispensable for anyone pursuing a career in ethical hacking. It provides a deep understanding of how networks operate, including knowledge of various protocols, IP addressing, subnetting, and the OSI and TCP/IP models. Mastering these fundamentals enables ethical hackers to identify and exploit vulnerabilities within network infrastructures effectively.

Additionally, such a course covers the concepts which are essential to understand security measures like firewalls, VPNs, and IDS/IPS. By honing skills in network monitoring and analysis, ethical hackers can better detect and mitigate potential threats.

Attending a networking training from CyberiumX will equip you with the critical knowledge and skills necessary to excel in ethical hacking, making you a valuable asset in the field of cybersecurity.

Course Duration- 20 Hours



NETWORKING COURSE SYLLABUS

Module 01- Introduction to Networking

1. **Basics of Networking**
2. **Purpose of Computer Networks**
3. **Types of Networks**
4. **Network Topologies**

Module 02- Networking Models

1. **Overview of the OSI Model**
2. **Detailed study of each layer**
3. **Importance of OSI Model**
4. **Overview of the TCP/IP Model**
5. **Comparison between OSI and TCP/IP Models**

Module 03- Network Protocols

1. **Fundamental Protocols**
2. **Transport Layer Protocols**
3. **Application Layer Protocols - HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS, DHCP**
4. **Port Numbers for Protocols**



Module 04- IP Addressing and Subnetting

1. IP Addressing
2. IPv4 and IPv6 Addressing
3. Public vs. Private IP Addresses
4. Static vs. Dynamic IP Addressing
5. Subnetting
6. Subnet Masks
7. Calculating Subnets and Subnetting
8. CIDR Notation

Module 05- Network Devices

1. Network Hardware Devices- Routers, Switches, Hubs, Bridges
2. End Devices- PC, Laptops, Phones, Servers

Module 06- : Configuration & Troubleshooting

1. Basic Network Configuration (IP Addressing, Gateway Settings)
2. VLANs and Inter-VLAN Routing
3. Common Network Issues
4. Troubleshooting
5. Tools and Techniques for Network Troubleshooting (ping, tracert, nslookup)

Module 07- Wireless Networking

1. Understanding Wireless Networks
2. Wireless Frequency
3. IEEE 802.11 Standards
4. ·WPA, WPA2, WPA3

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